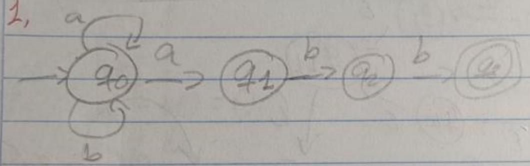
Tiffany Salazar Suarez 246?0

LABORATORIO 3

1.



$q_0 = \varepsilon + q_0 a + q_0 b = (a+b)^*$

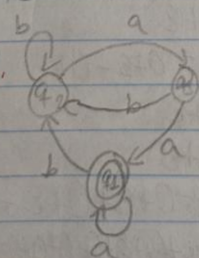
$q_1 = q_0\, a = (a+b)^* a$

$q_2 = q_1 b = (a+b)^* a b$

$q_3 = q_2 b = (a+b)^* abb$

$\therefore R_f = (a+b)^* abb$

2.



$q_1 = q_3 a + q_1 a = q_3 a a^*$

$q_2 = q_1 b + q_2 b + q_3 b = q_3 a a^* + q_2 b + q_3 b = q_3 (a a^* + b) + q_2 b$

$q_3 = q_2 a = q_3 (a a^* + b) b^* a$

$q_3 (a a^* + b) b^*$

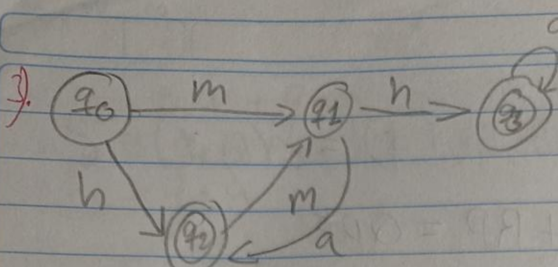
$q_3 = \varepsilon [(a a^* + b) b^* a]^*$

$q_1 = [(a a^* + b) b^* a]^* a + q_1 a$

$q_1 = [(a a^* + b) b^* a]^* a a^*$

$\therefore R_f = [(a a^* + b) b^* a]^* a a^*$

3.



$q_0 = \varepsilon$

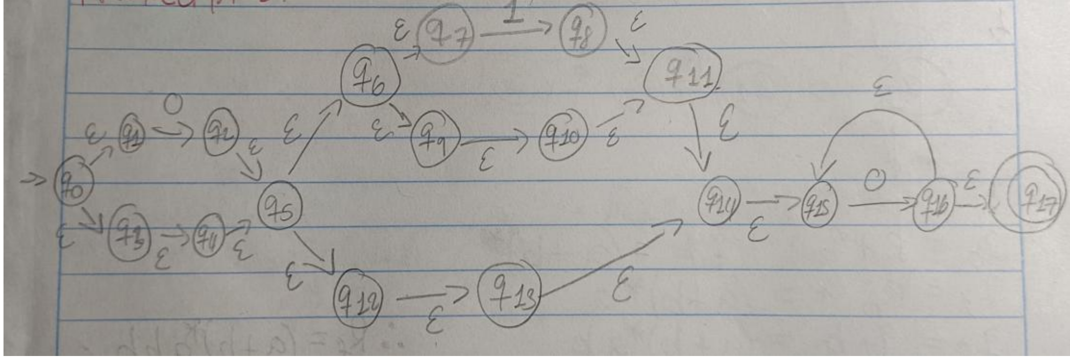
$q_1 = q_0 m + q_2 m = \varepsilon m + q_2 m = \varepsilon m + \varepsilon (h+m) a^*$

$q_2 = q_0 h + q_1 a = \varepsilon h + \varepsilon m + q_2 a = \varepsilon (h+m) a^*$

$q_3 = q_1 h + q_3 a = \varepsilon m + \varepsilon (h+m) a^* + q_3 a = \varepsilon (m+h+m) a^* a^*$

$\therefore R_f = (h+m) a^* (m+h+m) a^* a^*$

PROBLEMA 2:



$q_0 = \varepsilon$

$q_1 = \varepsilon q_0 = \varepsilon$

$q_2 = 0 q_1 = \varepsilon 0 = 0$

$q_3 = \varepsilon q_0 = \varepsilon$

$q_4 = \varepsilon q_3 = \varepsilon$

$q_5 = \varepsilon q_2 + \varepsilon q_4 = 0 + \varepsilon$

$q_6 = \varepsilon q_5 = 0 + \varepsilon$

$q_7 = \varepsilon q_6 = 0 + \varepsilon$

$q_8 = 1 q_7 = 1(0+\varepsilon)$

$q_9 = \varepsilon q_6 = 0 + \varepsilon$

$q_{10} = \varepsilon q_9 = 0 + \varepsilon$

$q_{11} = \varepsilon q_8 + \varepsilon q_{10} = 1(0+\varepsilon) + (0+\varepsilon)$

$q_{12} = \varepsilon q_5 = 0 + \varepsilon$

$q_{13} = \varepsilon q_{12} = 0 + \varepsilon$

$q_{14} = \varepsilon q_{11} + \varepsilon q_{14} = [1(0+\varepsilon)+(0+\varepsilon)] + \varepsilon q_{14} = [1(0+\varepsilon)+(0+\varepsilon)]\varepsilon^*$

$q_{15} = \varepsilon q_{14} + \varepsilon q_{16} = [1(0+\varepsilon)+(0+\varepsilon)]\varepsilon^* + 0 q_{15} = [1(0+\varepsilon)+(0+\varepsilon)]\varepsilon^* 0^*$

$q_{16} = 0 q_{15} = 0[1(0+\varepsilon)+(0+\varepsilon)]\varepsilon^* 0^*$

$q_{17} = \varepsilon q_{16} = 0[1(0+\varepsilon)+(0+\varepsilon)]\varepsilon^* 0^*$

$\therefore R_f = 0[(0+\varepsilon)(1+\varepsilon)]\varepsilon^* 0^*$

Link repositorio: https://github.com/Tiffany24630/Laboratorio_3-Teoria_de_la_computacion-/tree/problema1